



Wadhvani Ignite

CONCEPT BOOKLET

Module 5

Lesson 2 & 3: Build an Initial Prototype & Prototype for Early Validation

Context:

This lesson focuses on the critical steps involved in building an initial prototype, a key process for transforming startup ideas into tangible forms. Prototyping is essential for testing and validating concepts before committing significant resources. The lesson covers the different fidelity levels of prototypes, methods for creating prototypes for various types of products and services, and the specific steps to follow when building prototypes. Additionally, it outlines essential do's and don'ts to ensure prototypes effectively validate ideas and guide startups toward successful product development.

Program overview, and your **Venture Journey**

over 14 weeks

1. Problem Identification	2. Customer Identification & Needs Validation	3. Solution Idea Generation
Week 1-2	Weeks 2-4	Week 4
- Orientation - Develop Macro Industry-Problem view - Assemble "Real-World" Problems - Examine Chosen Problems - Analyze Problems	- Identify customer segments - Assess 'jobs-to-be-done' - Develop Customer Personas - Understand Customer Validation - Validate Customer-Problem fit - Pivot/Refine Customer-Problem Fit for Feasibility	- Understand Ideation techniques - Generate Solution Idea
4. Opportunity & Competition Mapping	5. Prototype Development & MVP	6. Opportunity Feasibility Assessment & Sizing
Week 5	Week 6	Week 7-8
- Identify Global competitors - Review Industry Trends and Estimate Market Size (TAM) - Introduction to Financial Planning	- Overview of Prototypes & MVP - Build an Initial Prototype - Prototype for Early Validation	- Map your Relative Position in the Market - Estimate Opportunity Size
7. Business Modelling	8. Marketing & Sales Strategy	9. Financial Management for Profitability
Week 8-9	Week 9-10	Week 11-12
- Examine Revenue Model - Revenue Models continued - Review and Organize the Lean Canvas - Build the LC for your Startup	- Understand Marketing & Sales - Recommend a Go-To-Market (GTM) Approach - Map a Sales Process	- Deep Dive into Financial Planning - Analyze the Path to Profitability - Understand Bootstrapping and Funding Options
10. Team & Talent Requirement	11. Orientation for Scale	12. Venture Idea Feasibility Presentation
Week 12	Week 13	Week 14
- Finding Co-founders and Mentors - Building an Initial Team	- Thinking Beyond the Prototype - Scale Opportunity & Orientation - Persuasive Story-Telling for a Viable Venture Idea - Pitch preparation	- Practice Pitch - Final Pitches

Concepts Covered in this Lesson:

Fidelity Levels of Prototype:

- Fidelity levels of Prototype
- Prototypes for different types of Products and Services
- Samples of Prototypes

Steps to take when building prototypes

- Do's and Don'ts: Prototypes and Validations

After completing this lesson, you will be able to:

- 1. Understand the different fidelity levels of prototypes and choose the appropriate level for your needs.**
- 2. Create prototypes for various types of products and services, including software, physical products, and services.**
- 3. Follow a structured process to build effective prototypes.**
- 4. Identify and use the right tools and materials for prototyping.**

Steps for you to take:

Identify Prototype Fidelity Level:

Determine the appropriate fidelity level for your prototype based on your current development stage.

Create and Test a Prototype:

Build an initial prototype using the chosen fidelity level and test it with users or stakeholders.

Fidelity Levels of Prototypes

Prototypes can vary in their level of detail and functionality, known as fidelity. Understanding these different levels helps in choosing the right type of prototype for various stages of product development. The three main fidelity levels are low, medium, and high, each serving different purposes and providing varying degrees of insight and feedback.

Low-Fidelity Prototypes

These prototypes are basic and simplified representations of the product. They focus on the core idea rather than detailed design or functionality.

Examples: Sketches, paper models, and rough digital mock-ups.

Purpose: Ideal for early-stage brainstorming and concept validation. They are quick and inexpensive to produce, allowing for rapid iteration and broad exploration of ideas.

Medium-Fidelity Prototypes

More detailed and interactive than low-fidelity prototypes, providing a clearer representation of the product's design and some functionality.

Examples: Wireframes, detailed mock-ups, and basic 3D models.

Purpose: Useful for testing specific functionalities and gathering user feedback. They help in refining the product's design and understanding how users interact with it, without the complexity and cost of high-fidelity prototypes.

High-Fidelity Prototypes

These prototypes closely resemble the final product in both appearance and functionality. They are detailed, polished, and often fully functional.

Examples: Working models, fully functional software, and detailed physical models.

Purpose: Used for final validation and user testing before production. They provide a realistic experience, allowing for thorough testing and refinement. High-fidelity prototypes are essential for gathering detailed feedback and ensuring that the product meets user expectations and requirements.

Prototypes for different types of Products and Services

Prototypes are crucial for testing and refining ideas across various types of products and services. Depending on the nature of the product or service, different prototyping methods are applicable. This section explores how to create prototypes for software, physical products, and services, providing relevant examples and methods for each.

Software Prototypes

Software prototypes are digital representations of applications, websites, or software solutions. They can range from simple wireframes to interactive, clickable mock-ups.

Examples:

Wireframes: Basic layout sketches showing the structure of the application.

Mock-ups: More detailed visual representations with design elements.

Clickable Prototypes: Interactive models that simulate user interactions.

Methods:

Use tools like **Sketch**, **Figma**, or **Adobe XD** for creating wireframes and mock-ups.

Utilize prototyping platforms such as **InVision** or **Axure** for building interactive prototypes.

Conduct usability testing with potential users to gather feedback and refine the software.

Product Prototypes

Product prototypes are physical representations of tangible items. They can range from simple models to fully functional versions.

Examples:

3D Printed Models: Basic forms to test design and ergonomics.

Functional Prototypes: Working models that demonstrate the product's features and functionality.

Appearance Models: Highly detailed models that resemble the final product in look and feel but may not be functional.

Methods:

Use **3D printing** for rapid prototyping and testing form and fit.

Create functional prototypes using materials and components similar to those of the final product.

Develop appearance models for marketing and user feedback using techniques like **CNC machining** or **vacuum casting**.

Service Prototypes

Service prototypes simulate the delivery and interaction of service offerings. They help in visualizing and testing the service processes and customer experiences.

Examples:

Role-Playing: Acting out the service interactions to understand customer experience.

Storyboards: Visual sequences showing the steps and interactions in the service process.

Service Blueprints: Detailed diagrams mapping out the service delivery process, including customer touchpoints and backstage operations.

Methods:

Conduct **role-playing** sessions with team members to identify potential issues and improvements.

Create **storyboards** to visualize and communicate the service process.

Develop **service blueprints** to map out the entire service delivery and identify areas for optimization.

Samples of Prototypes

Samples: Prototypes for Software Applications

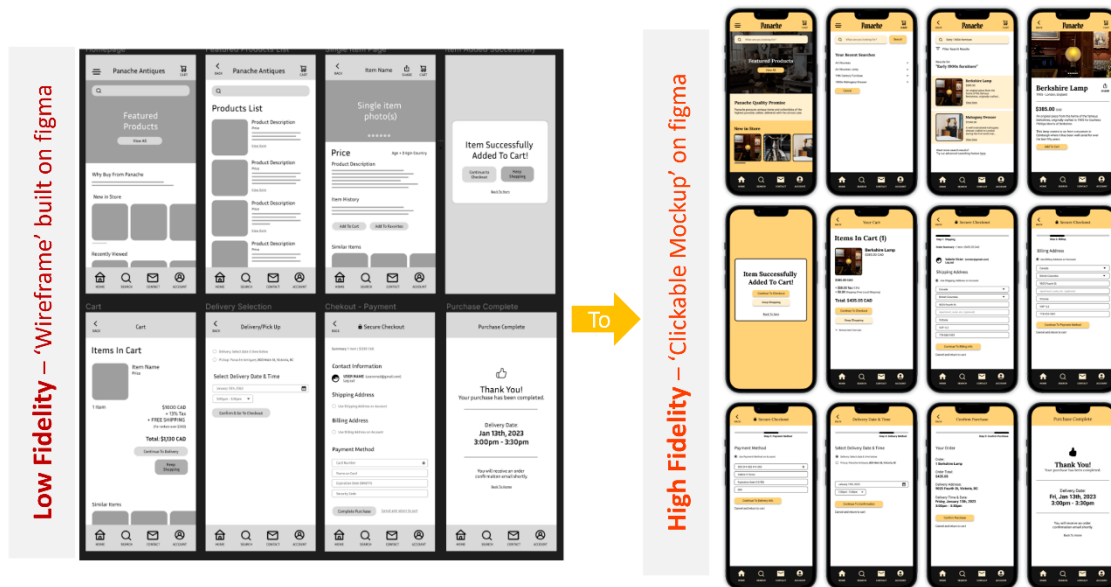
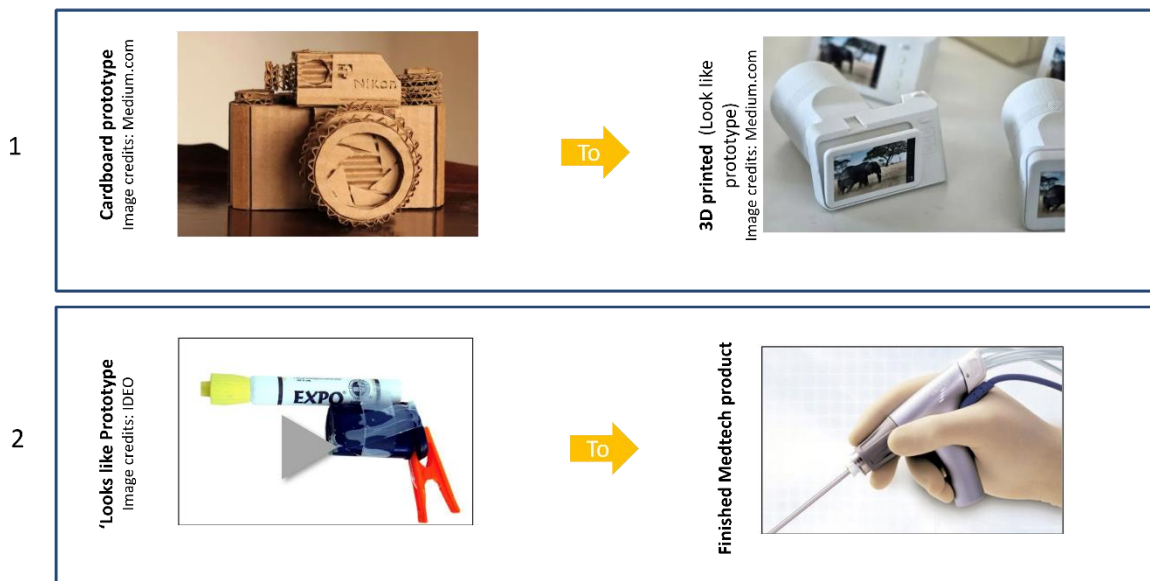


Image credit: Sarah, Lessonsindesign.com

Samples: Prototypes for 'Product' (manufacturing)



Samples: Prototypes for Services

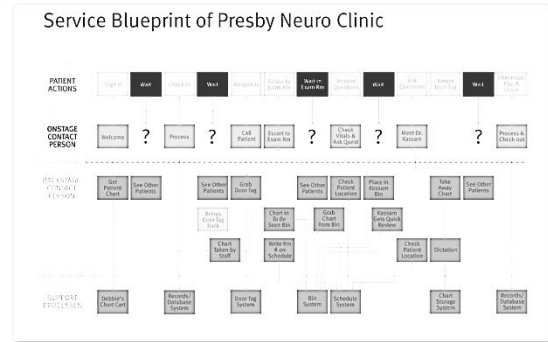
Low Fidelity – 'Services staging Role Play'



"It is a kind of theatre rehearsal of the service in the real-life surroundings. The design team with shareholders or even potential customers take part in it. Due to the "safe" environment all the participants feel free and open enough for pure experimentation. They are invited to change roles from the customer to service provider and follow the different scenarios. Every scenario may be iteratively played by each pair of participants."



High Fidelity – 'Blueprints'



"Services blueprint is the most detailed visualization tool of services testing...it details all the points from the customer contact to the backstage processes and consolidate them. As an outcome, we have a detailed visualization of all the interactions and processes, without any emotional component."

Source: Startus Magazine – 'Design Thinking Prototypes to Role play'

Steps to take when building prototypes

The steps involved in building prototypes vary depending on whether the focus is on software, physical products, or services.

Software Prototypes

Determine User Interactions:

Focus on how users will interact with the software.
Do not concentrate on databases or back-end designs yet.

Create Wireframes and Mockups:

Use tools like Balsamiq, Sketch, or Figma to design the interface.
Develop visual representations of user flows and screen layouts.

Build Interactive Prototypes:

Utilize tools such as Bubble, Adobe XD, InVision, or Axure.
Create interactive prototypes to simulate user interactions.

Test the Prototype:

Conduct usability testing with target users.
Gather feedback on user experience and interface design.

Iterate and Refine:

Make adjustments based on feedback.
Retest to ensure improvements meet user needs.

Product Prototype

Start with Simple Design Sketches:

Visualize the product using paper or digital drawing tools.
Create basic sketches to outline the product's design and features.
Create 3D Models:

Use software like SolidWorks, Tinkercad, or Autodesk Fusion to develop detailed 3D models.
Ensure the 3D model captures all critical aspects of the product's design.

Build a Physical Representation:

Use materials such as cardboard, foam, or 3D printing to create a physical prototype.

Focus on replicating the form, fit, and basic functionality of the product.

Test the Prototype:

Evaluate the design, ergonomics, and functionality with users and stakeholders.

Collect feedback on usability and overall design.

Iterate and Refine:

Make necessary adjustments based on feedback.
Create new iterations as needed to improve the prototype.

Service Prototypes

Create a Service Blueprint:

Outline the entire customer journey.
Detail frontstage and backstage actions, and touchpoints involved in the service.

Develop a Customer Journey Map:

Create a graphical representation of the customer's experience.

Detail each touchpoint and the emotional state of the customer throughout the service.

Role-Play the Service:

Simulate the service experience to understand and improve the customer experience.

Use storytelling techniques to visualize and enhance service delivery.

Test the Service Prototype:

Conduct simulations or pilot programs with real users.
Collect feedback on customer experience and service delivery.

Iterate and Refine:

Make necessary adjustments based on feedback.
Continuously improve the service prototype through repeated testing and refinement.

Do's and Don'ts: Prototypes and Validations

Prototyping and validation are crucial steps, help in refining ideas, identifying potential issues, and ensuring that the final product meets user needs and expectations. Understanding the best practices and common pitfalls associated with prototyping and validation can significantly improve the effectiveness of these processes. The following guidelines provide essential do's and don'ts to help you navigate the complexities of prototyping and validation successfully.

Prototyping: Do's & Don'ts

Do's	Don'ts
Keep it quick and dirty	Don't over-engineer: Don't fall into the trap of making your prototype too complex
Get the team aligned on the future final solution	Don't be afraid to fail: Not every prototype will be a success. That's okay!
Think of your prototype as a living document, not a finished product	Don't get attached to your prototype
A good prototype should communicate the vision and potential of your product	Don't confuse prototypes with the final product
Bells and whistles come later. Keep it simple and quick for now	Don't forget: Its purpose is to learn, not launch

Prototype Validation: Do's & Don'ts

Do's

Ensure your testing group represents your target audience

Start testing as early as possible and continue throughout the development process

Encourage honest feedback and be open to making changes based on user suggestions and needs.

Prioritize testing the core features and functionalities that are crucial to your product's success.

Keep detailed records of test results, user feedback, and changes made.

Don'ts

Don't Ignore Negative Feedback: Negative feedback is as valuable as positive feedback, if not more so.

Don't Test with Too Narrow a Group: Ensure your test group is diverse enough to represent your entire target audience.

Don't Rush the Process: Take the time needed to thoroughly test and validate your prototype.

Don't Overcomplicate Your Prototype: Keep your prototype focused and simple. Adding too many features too early can complicate testing and feedback.

Call-To-Action:

Complete Venture Activity 5.1: Build Your Prototype, 5.2 Collect Feedback on Prototype and 5.3 Analyze Insights from Prototype Validation.

Determine Fidelity Levels:

Assess the different fidelity levels of prototypes.

Choose the appropriate fidelity level for your prototype.

Create Prototypes:

Develop prototypes for software, physical products, and services.

Use suitable tools and methods to create each type of prototype.

Identify and Use Tools and Materials:

Select the right tools and materials needed for your prototypes.

Gather and utilize these resources effectively.

Complete Prototype Validation:

Prepare your prototypes for validation testing.

Conduct validation tests with relevant stakeholders and gather feedback.

Analyse the results and iterate on your prototypes based on the feedback received.